

ROBBINS

Soft-Rigid Robotic Joints Controlled By Deep Reinforcement-Learning

MGMT 162: Team 6 Technology Commercialization Deck

Try Pitch

Team



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National Library of Medicine
National Center for Biotechnology Information

In recent years technical innovations have combined to make artificial limbs much more comfortable, efficient, and lifelike than earlier versions. Future innovations are likely to depend on the interaction between three powerful forces—amputees' demands, advances in surgery and engineering, and healthcare funding sufficient to sustain development and application of technological solutions. This article looks at the innovative new prostheses that are currently available and discusses future developments.

Artificial Limbs Market to Grow at 5.1% CAGR, Crossing $3.5 Billion by 2031: TMR Study

Wilmington, Delaware, United States, Aug. 03, 2023 (GLOBE NEWSWIRE) -- The global **artificial limbs market** was valued at US$ 2.3 Bn in 2022 and is projected to reach US$ 3.5 Bn by 2031.

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01

Technology Overview

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1. Soft-Rigid Structure

2. Nonlinear Control

3. Deep Reinforcement-Learning

4. Additional features

Background

What motivates our invention?

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1. Soft-Rigid Structure

2. Nonlinear Control

3. Deep Reinforcement-Learning

4. Additional features

Organic Imitation

The combination of elasticity and rigid structure found in organic limbs allows for flexibile structure, versatile behavior, and wide range of motion

However...

Current prosthetic limbs cannot yet fully replicate organic imbs, and most modern robots are either primarily soft or rigid

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1. Soft-Rigid Structure

2. Nonlinear Control

3. Deep Reinforcement-Learning

4. Additional features

Linear Controller

Rigid structure have fixed degree of freedom, which linear controller, such as proportional integral derivative (PID) controller, can control it effectively

New Problem

The introduction of flexiblity in the structure creates nonlinear dynamic, which requires an addition of nonlinear controller to control it effectively

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1. Soft-Rigid Structure

2. Nonlinear Control

3. Deep Reinforcement-Learning

4. Additional features

Solution

Machine learning, specifically reinforcement-learning and deep reinforcement-learning, can be used to implement nonlinear controller

Machine Learning

Reinforcement learning (RL) solves the problems through a data driven approach. Deep reinforcement learning (DRL) extends this framework by using their own results.

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1. Soft-Rigid Structure

2. Nonlinear Control

3. Deep Reinforcement-Learning

4. Additional features

Strength

Soft-rigid structure allows for the joints to handle external forces better without breaking

Safety

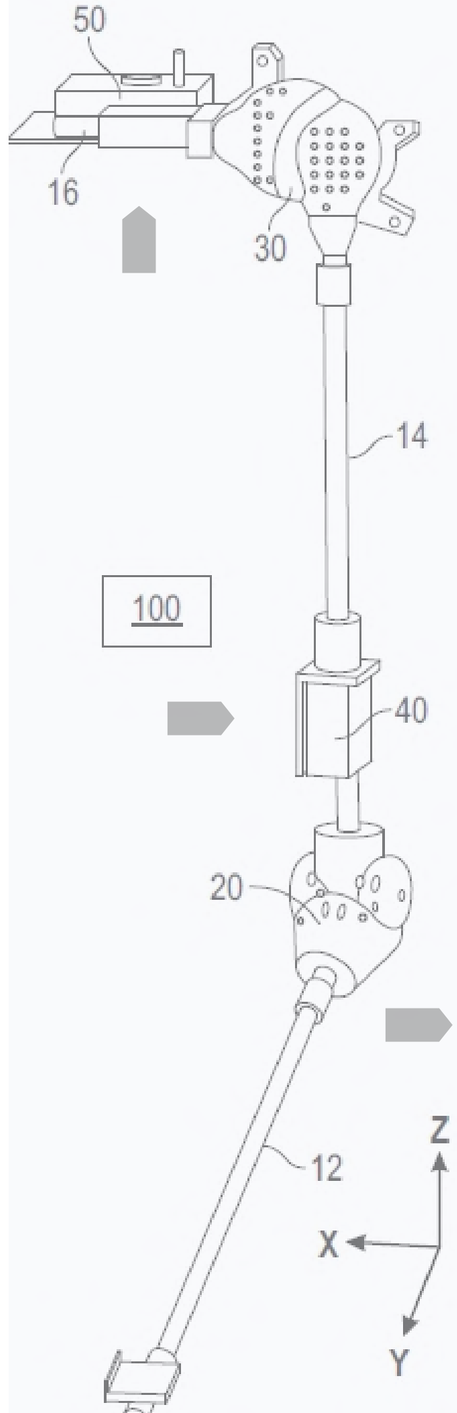
Flexibility in soft-rigid structure mitigates forces on contact when mistakenly striking humans

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INTRODUCING...

Robbins

Soft-Rigid Robotic Joints Controlled By Deep Reinforcement-Learning



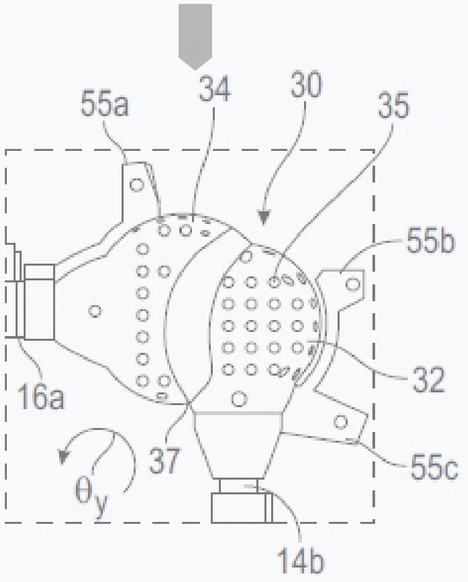


FIG. 3

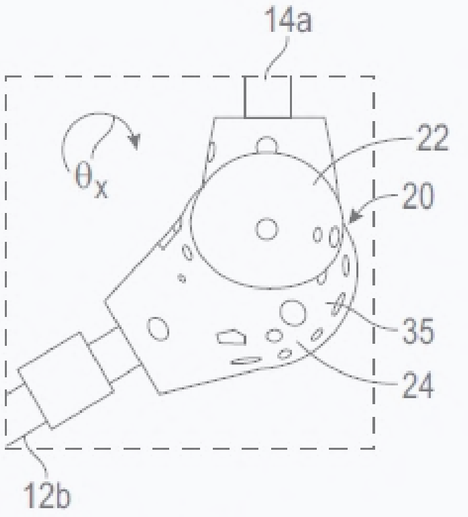


FIG. 4

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Through-holes

introduce flexibility to the structure, which can be adjusted through the modification of hole quantity, size, and material (such as silicone).

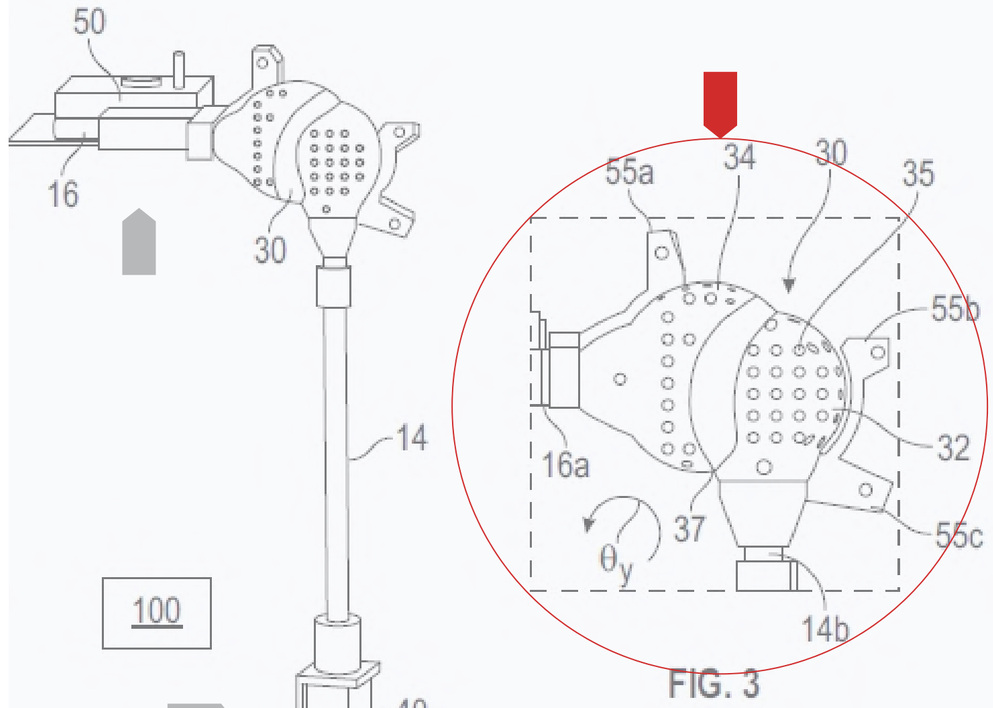


FIG. 3

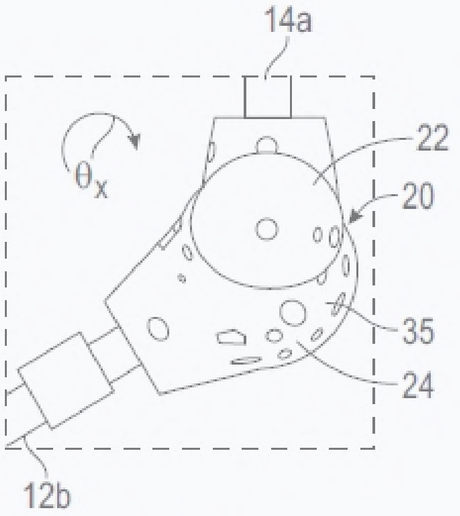


FIG. 4

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Elastic Membrane

The external elastic membrane provides a base of elasticity that gets compressed and stretched on opposite sides when the joint is opening or closing.

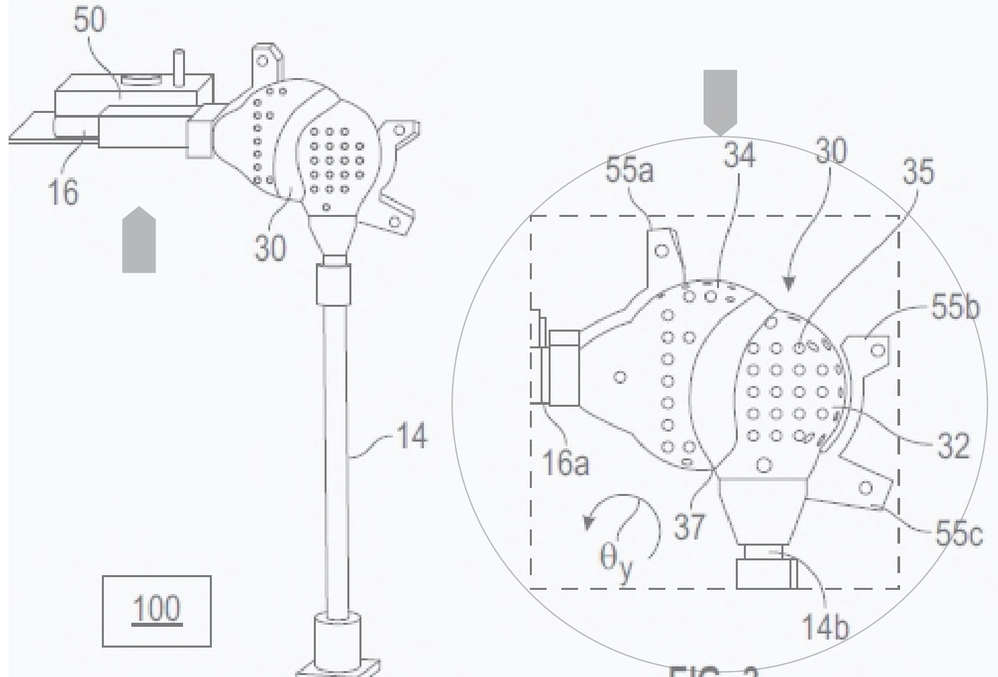


FIG. 3

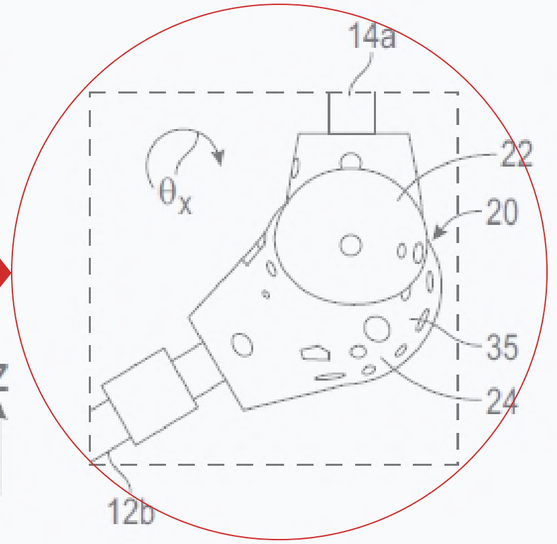


FIG. 4

Try Pitch

Deep reinforcement learning (DRL)

Technology that involves nonlinear dynamics and controllable through machine learning methods

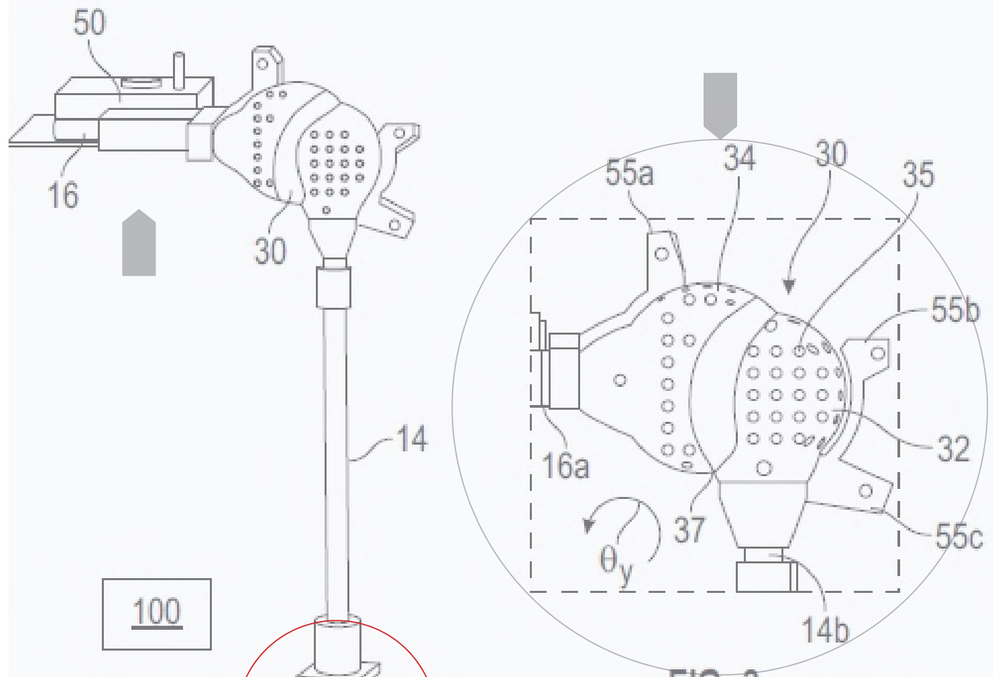


FIG. 3

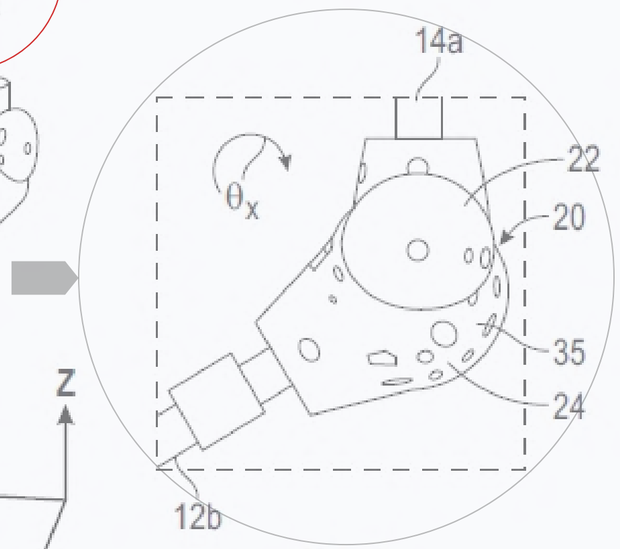
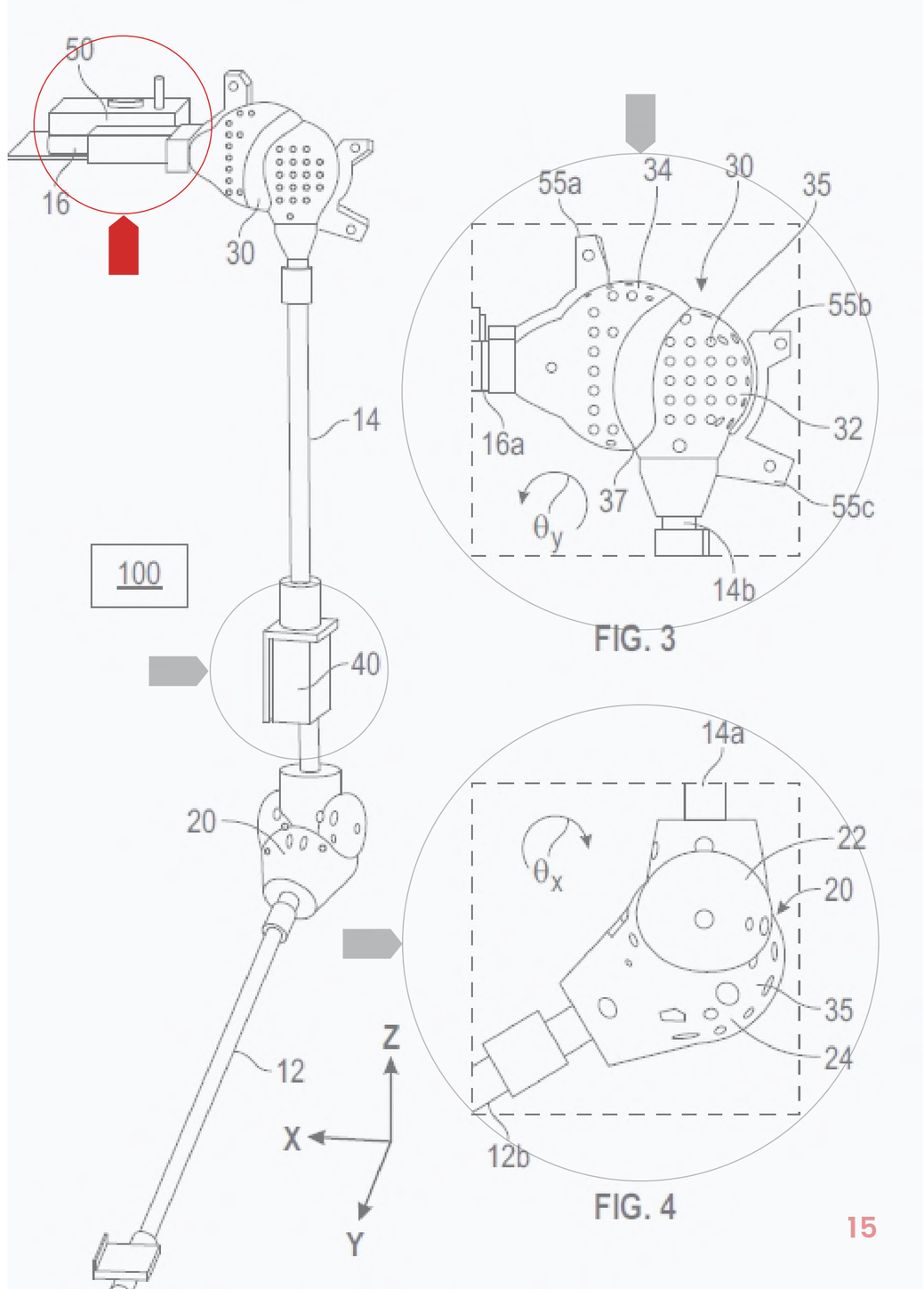


FIG. 4

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Soft Actor-Critic (SAC) algorithm

Model-free control methodology that yields grater control precision than PID controller alone.



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Intellectual Property Assessment

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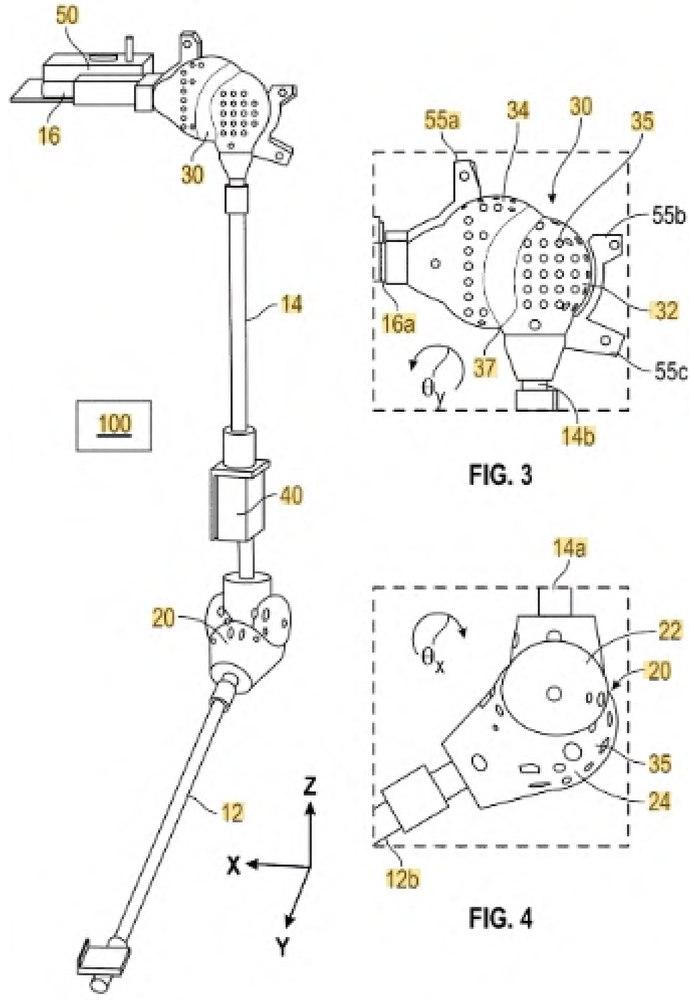


FIG. 2

FIG. 3

FIG. 4

IP ASSESSMENT

Our Patent

Soft-rigid robotic joints controlled by deep reinforcement-learning

- US20220281123A1
- University of California, Santa Cruz

  PATENT PENDING

  Filed on 2022-03-02

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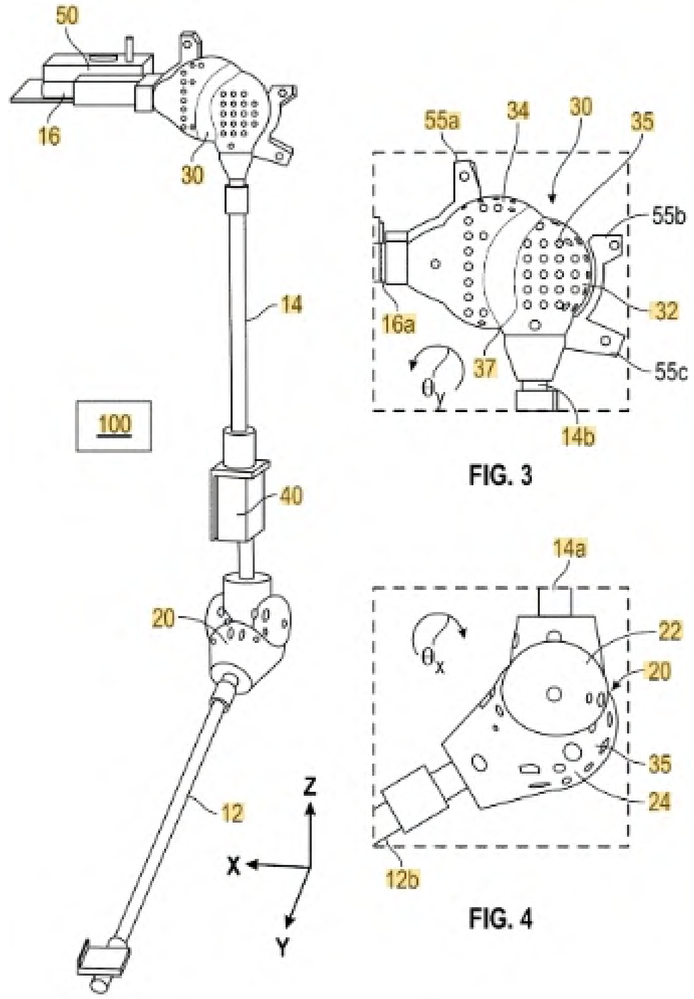


FIG. 2

FIG. 3

FIG. 4

IP ASSESSMENT

Our Claims

Soft-rigid robotic joints controlled by deep reinforcement-learning

- Robotic arms, through-holes and sockets
- Flexible membrane
- A method of manufacturing a robotic joint
- Programming control agent to control a robotic arm

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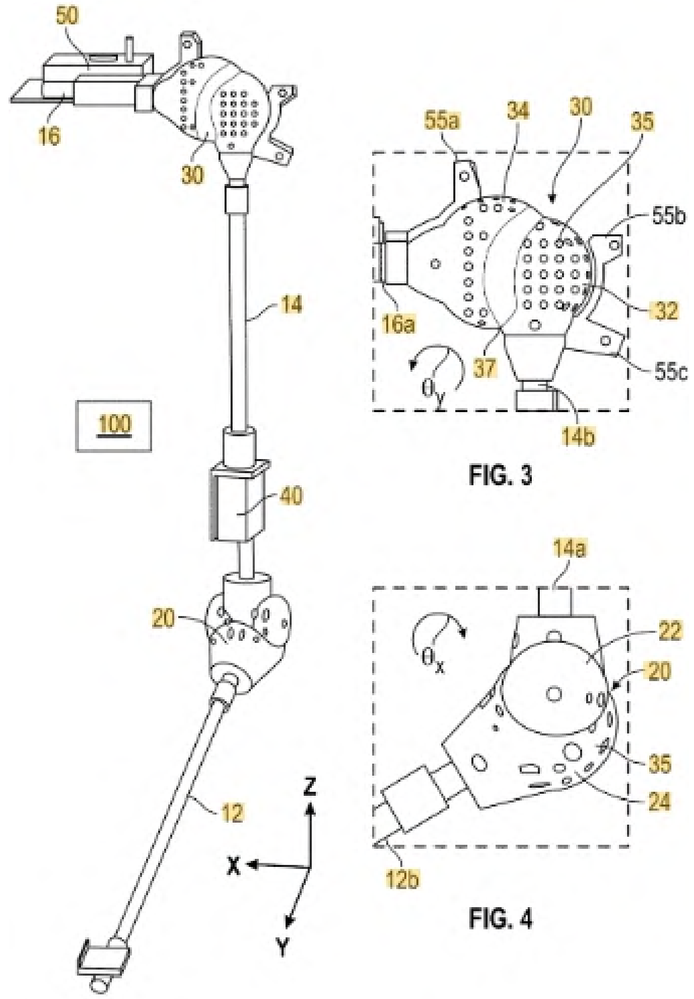


FIG. 2

FIG. 3

FIG. 4

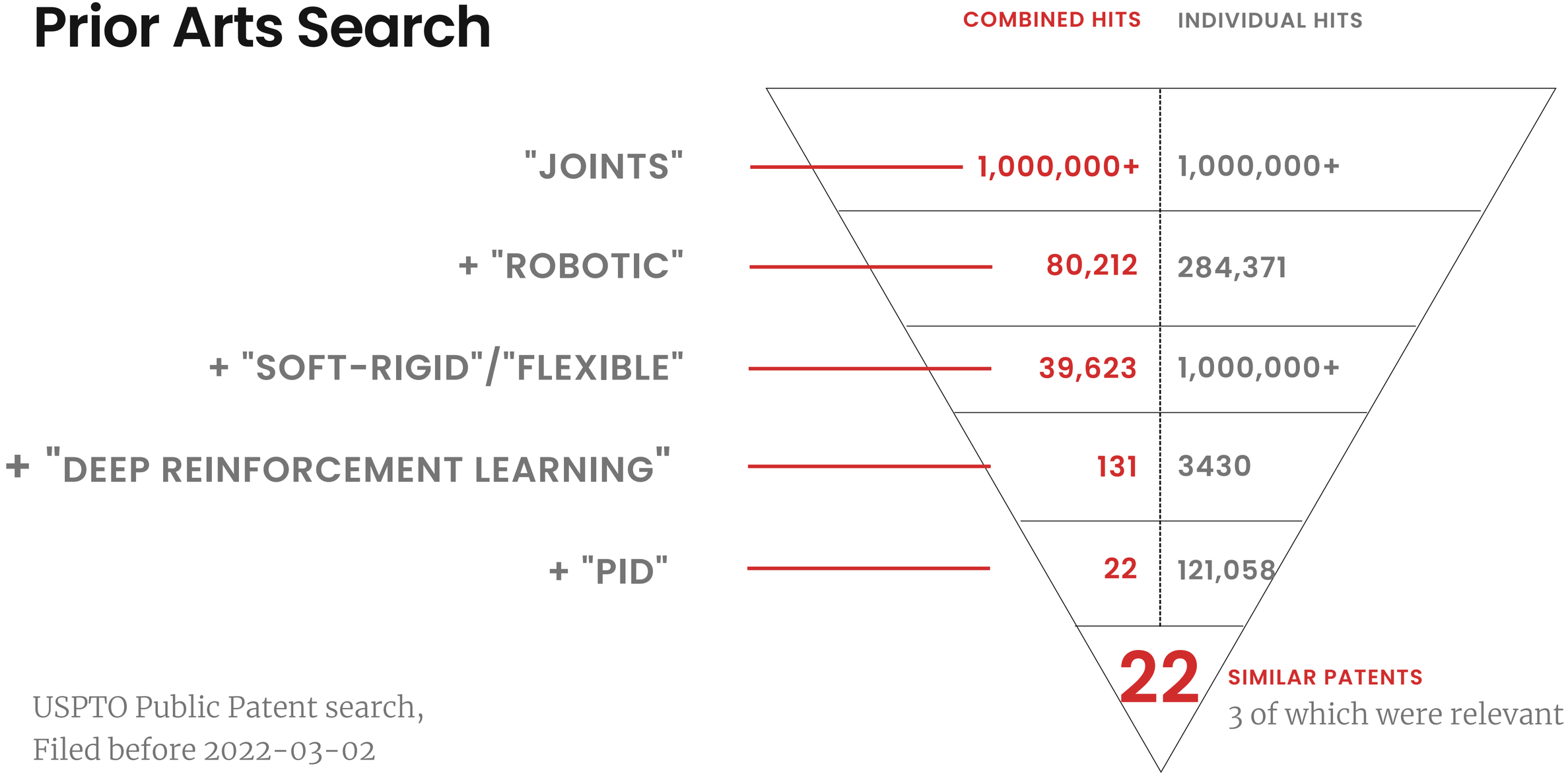
IP ASSESSMENT

Codes

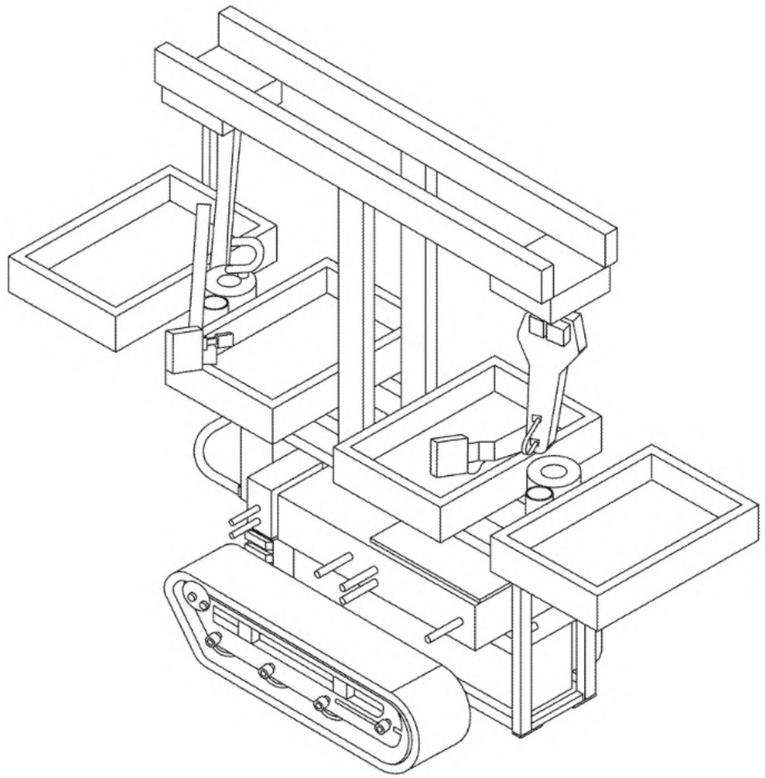
- **B25J17/00:** Joints
- **B25J17/0275:** Universal joints...
- **B25J9/163:** Programme controls characterised by the control loop learning...
- **B25J9/1641:** Programme controls characterised by the control loop compensation...
- **B25J9/1671:** Programme controls characterised by programming...
- **G05B2219/33056:** Reinforcement learning...
- **G05B2219/39186:** Flexible joint

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Prior Arts Search



USPTO Public Patent search,
Filed before 2022-03-02



PRIOR ARTS

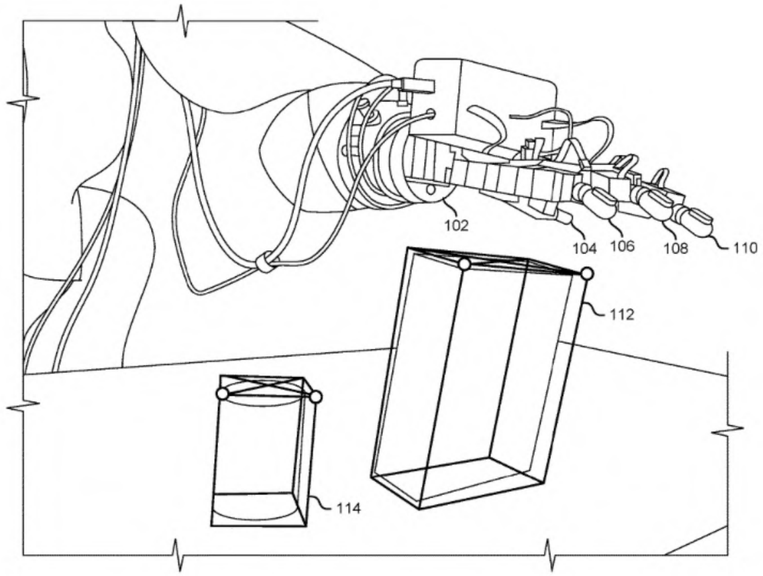
Robotic Fruit Picking System

- US 20190261565 A1
- Dogtooth Technologies Ltd.
- Filed on 2019-05-08

Noteable Claims:

- Picking arm with flexible picking head
- Reinforcement learning approach
- Physical machine for picking fruit

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PRIOR ARTS

Robotic Control System

- US 20200061811 A1
- **NVIDIA** Corp.
- Filed on 2019-08-23

Noteable Claims:

- Arm with gripper to grab 3D object
- Uses a machine learning trained using double-deep reinforcement learning (“DRL”)
- Grabs object to maximize grip

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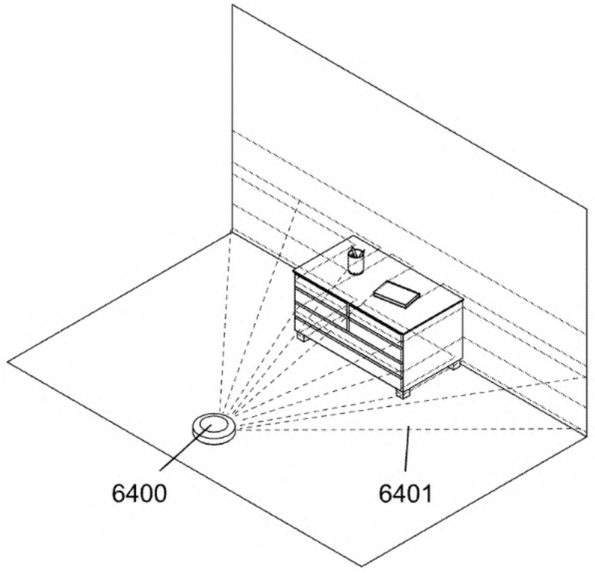
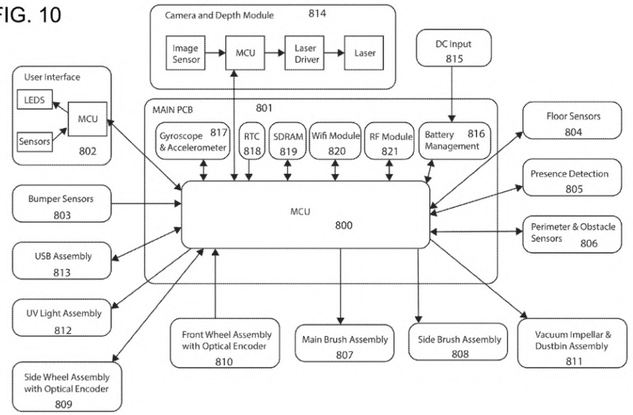


FIG. 10



PRIOR ARTS

Obstacle Recognition Method for Autonomous Robots

- US 20220066456 A1
- AI Inc.
- Filed on 2021-08-16

Noteable Claims:

- Comparing object from the captured images to objects in an object dictionary
- Actuating the robot to drive along a planned trajectory

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IP ASSESSMENT

Summary

Novelty

- silicone with soft-rigid
- AND reinforcement learning to train flexible robotic arm

Non-obviousness

- The through-holes within the mechanism is rare
- Uses deep reinforcement learning is not obvious

Usefulness

- Improves compliance for control and safety
- Maintains precision
- Is more biologically analogous to real arm

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03

Market Analysis

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Market Size

Total Addressable Market (TAM)

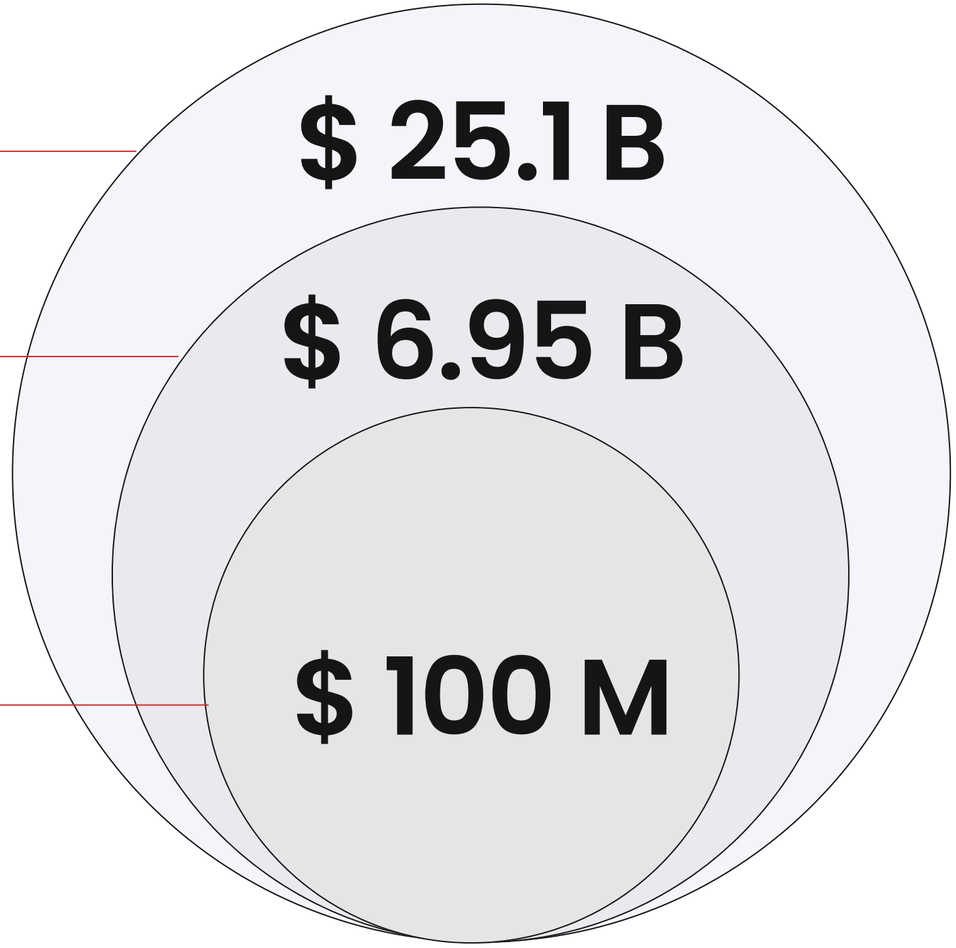
ROBOTICS

Serviceable Available Market (SAM)

ROBOTIC PROSTHETICS

Serviceable Obtainable Market (SOM)

1.5% OF ROBOTIC PROSTHETICS



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Competitors Overview

ottobock.

innovative solutions such as the C-Leg microprocessor knee and Genium X3 which significantly enhance the mobility and quality of life for users.

Focus Areas: Prosthetics, Orthotics, Mobility solutions (wheelchairs), Exoskeletons

ÖSSUR

advanced bionic limbs like the Rheo Knee and Proprio Foot, integrating cutting-edge technology to provide superior functionality and comfort.

Focus Areas: Bionic limbs, Non-invasive bracing solutions

Blatchford

lightweight, energy-efficient, with flagship products - Echelon foot and Linx integrated limb system, focusing on natural movement and high user satisfaction.

Focus Areas: Lower-limb prosthetics, Intelligent control systems, Energy-efficient designs

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Comparative Analysis

Factor	Össur	Ottobock	Blatchford Mobility	Robbins
Technology & Innovation	Advanced prosthetics with microprocessor control	Comprehensive prosthetic and orthotic solutions	Lightweight, energy-efficient prosthetics	Hybrid soft-rigid joints with DRL control
Product Portfolio	Bionic limbs, braces	Prosthetics, orthotics, mobility solutions	Lower-limb prosthetics	Robotics, healthcare, manufacturing
Customization	High customization	Custom solutions tailored to users	Intelligent, adaptive systems	Real-time adaptation with DRL
Market Reach	Global presence	Extensive global network	Primarily UK and Europe, expanding globally	Emerging, focusing on partnerships and licensing
R&D Focus	Continuous innovation	Multiple global innovation centers	Lightweight materials, biomechanics	Hybrid design and machine learning
User Experience	High user satisfaction	Positive feedback for advanced tech	Natural movement, energy efficiency	Improved safety, flexibility, and control
Cost & Accessibility	Premium pricing	Range of price points	Competitive pricing	Reduced costs through 3D printing

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04

Conclusion & Recommendation

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SWOT Analysis

S *W* *O* *T*

STRENGTHS

- Innovative Technology
- Cost-Effective Manufacturing
- High Precision and Safety

WEAKNESSES

- Complexity of Implementation
- Costumers don't care about the technology used as long as its working
- Limited Revenue Streams

OPPORTUNITIES

- Growing market size
- Technological trends
- Strategic Partnerships

THREATS

- Very high Competition
- Regulatory Challenges

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Robbins is NOT ready for commercialization

- High Competition
- No strong Differentiator
- Competitive Parity Risk

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OTHER MARKETS TO CONSIDER

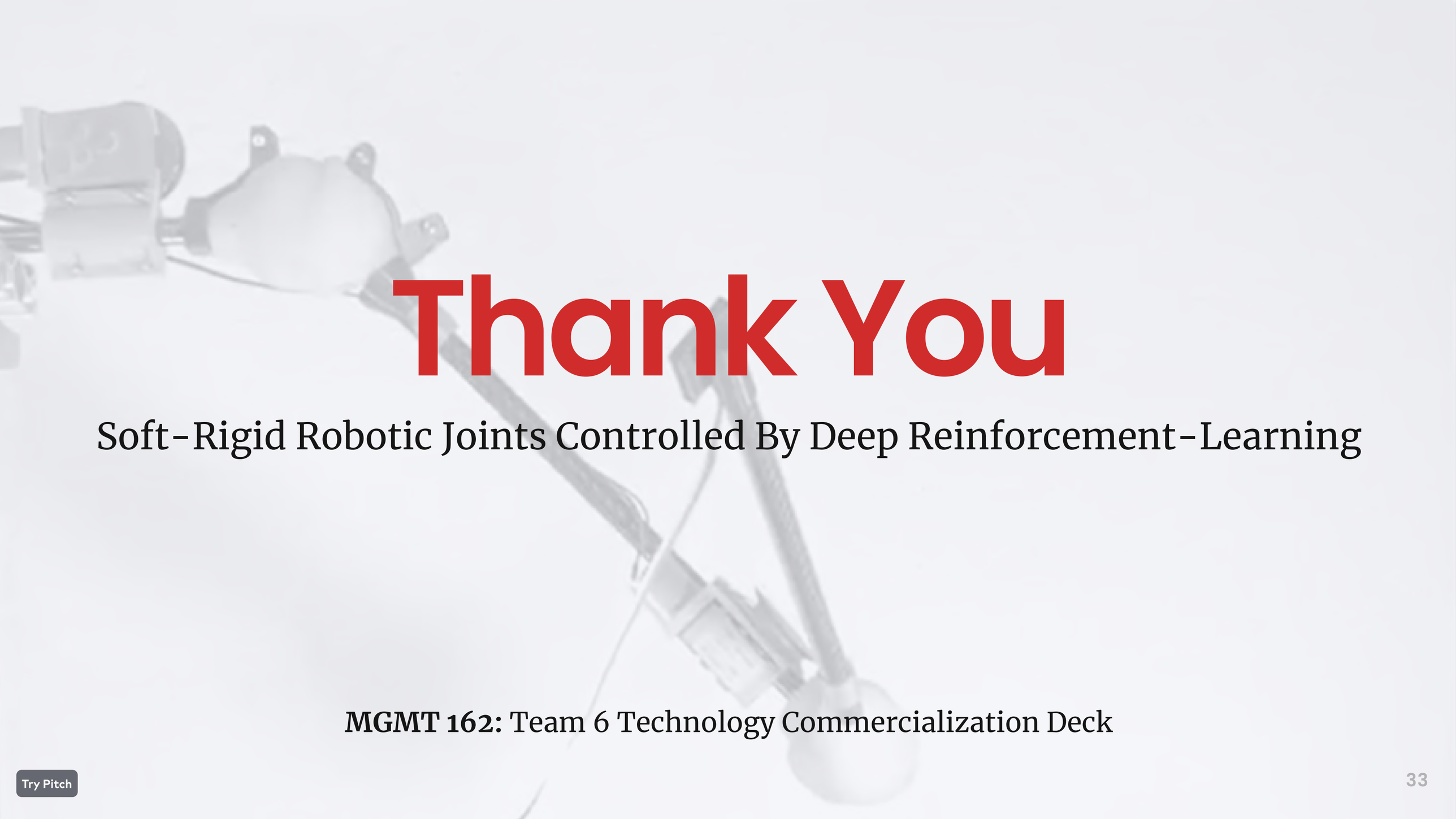
1 Healthcare Robots- surgical, rehabilitation

2 Manufacturing & Industrial Automation

3 Domestic Robots



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Thank You

Soft-Rigid Robotic Joints Controlled By Deep Reinforcement-Learning

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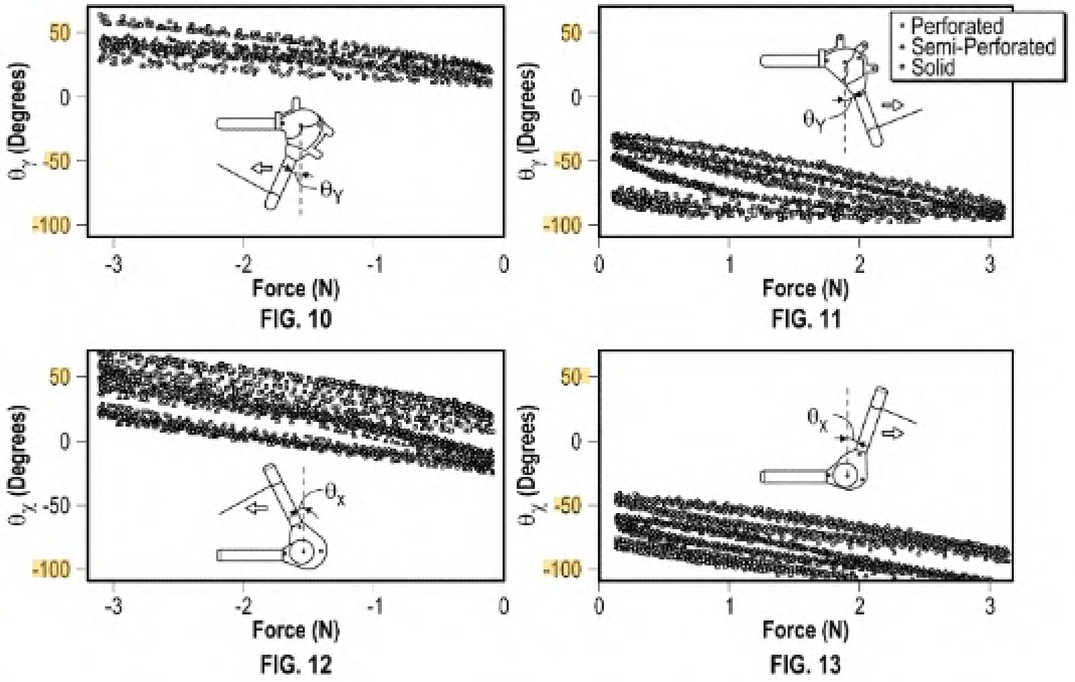
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05

Appendix

Appendix 1.1 - Force of Motion Measurement

- FIG. 10 is a plot of force measured as a function of an angle during closing of the shoulder joint according to the present disclosure
- FIG. 11 is a plot of force measured as a function of an angle during opening of the shoulder joint according to the present disclosure
- FIG. 12 is a plot of force measured as a function of an angle during closing of the elbow joint according to the present disclosure
- FIG. 13 is a plot of force measured as a function of an angle during opening of the elbow joint according to the present disclosure



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Appendix 1.2 - Drop Test

- FIG. 14 are plots (angle as a function of time) of drop test performed on robotic arms having a shoulder joint with a ball component being one of the three types (perforated, semi-perforated, solid) according to the present disclosure;
- FIG. 15 are plots (angle as a function of time) of a drop test performed on a robotic arm having a shoulder joint with a perforated ball component, a simulated drop test, and an idealized drop test, according to the present disclosure
- FIG. 16 are plots (angle as a function of time) of drop test performed on robotic arms having an elbow joint with a ball component being one of the three types (perforated, semi-perforated, solid) according to the present disclosure
- FIG. 17 are plots (angle as a function of time) of a drop test performed on a robotic arm having an elbow joint with a perforated ball component, a simulated drop test, and an idealized drop test, according to the present disclosure

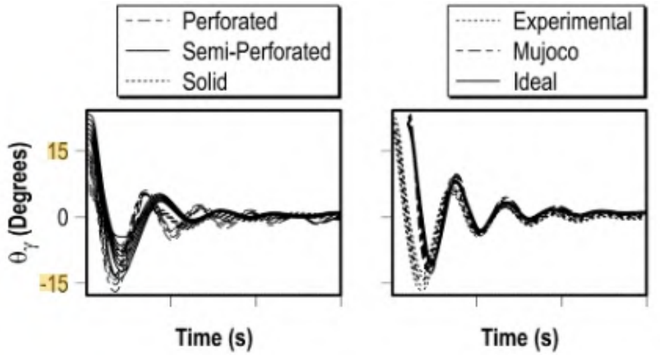


FIG. 14

FIG. 15

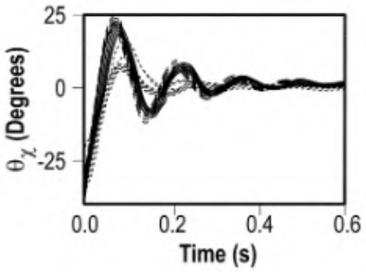


FIG. 16

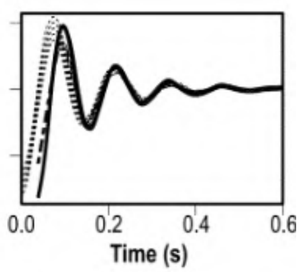


FIG. 17

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Appendix 1.3 - 3D Visualization

- FIG. 18 shows a 3D visualization of an end effector trajectory moved by a robotic arm according to the present disclosure
- FIG. 19 shows a plot of an angle of the shoulder joint as a function of time of the robotic arm moving the end effector of FIG. 18
- FIG. 20 shows a plot of an angle of the elbow joint as a function of time of the robotic arm moving the end effector of FIG. 18

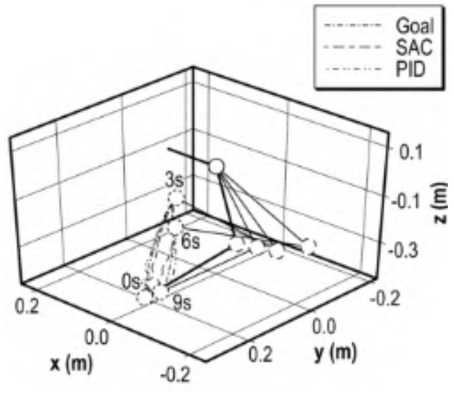


FIG. 18

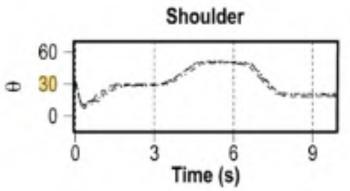


FIG. 19

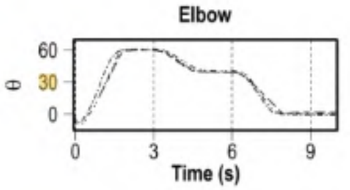


FIG. 20

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Appendix 1.4 - Joint Test

- FIG. 21 shows plots of a measured angle of the shoulder joint while performing an increasing steps maneuver using a SAC algorithm and a PID algorithm and the resulting error for each
- FIG. 22 shows plots of a measured angle of the elbow joint while performing an increasing steps maneuver using a SAC algorithm and a PID algorithm and the resulting error for each
- FIG. 23 shows plots of a measured angle of the shoulder joint while performing a decreasing steps maneuver using a SAC algorithm and a PID algorithm and the resulting error for each;

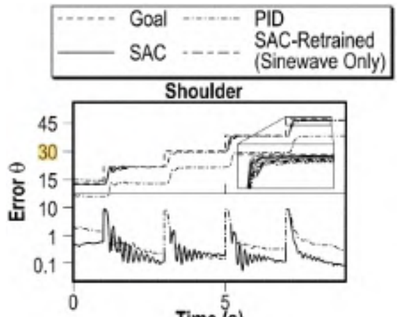


FIG. 21

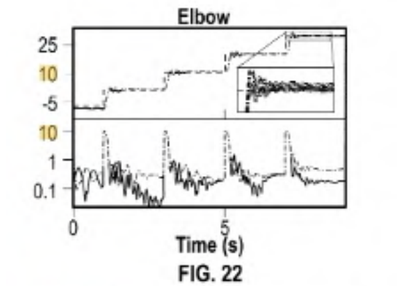


FIG. 22

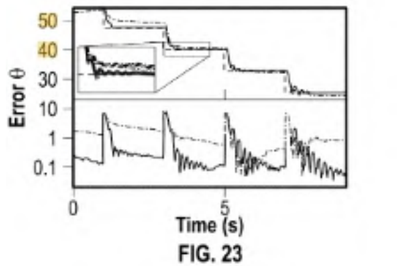


FIG. 23

Appendix 1.5 - SAC Algorithm Test

- FIG. 24 shows plots of a measured angle of the elbow joint while performing a decreasing steps maneuver using a SAC algorithm and a PID algorithm and the resulting error for each
- FIGS. 25 and 26 show goal (ideal) plots of a measured angle of the shoulder joint while performing a sine wave maneuver along with plots using a SAC algorithm, a PID algorithm, a retrained SAC algorithm and the resulting error for each;

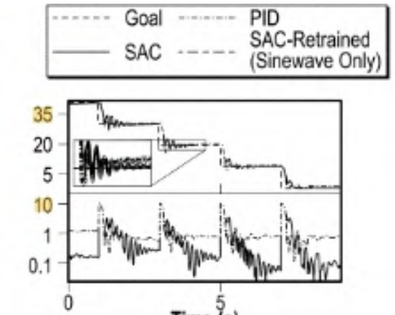


FIG. 24

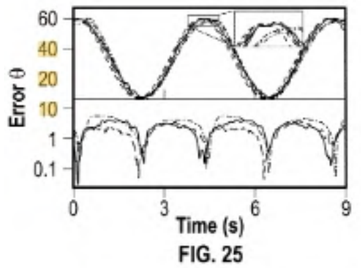


FIG. 25

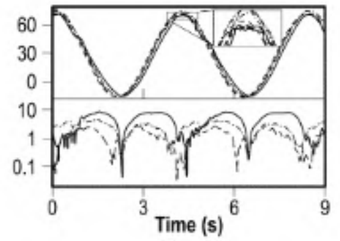


FIG. 26

Appendix 1.6 – Error Plots

- FIG. 27 shows absolute error plots as a function of time for each of three types of joint components, for each of the shoulder joint, the elbow joint, and the robotic arm, performing three motions (increasing steps, decreasing steps, and sinus wave);

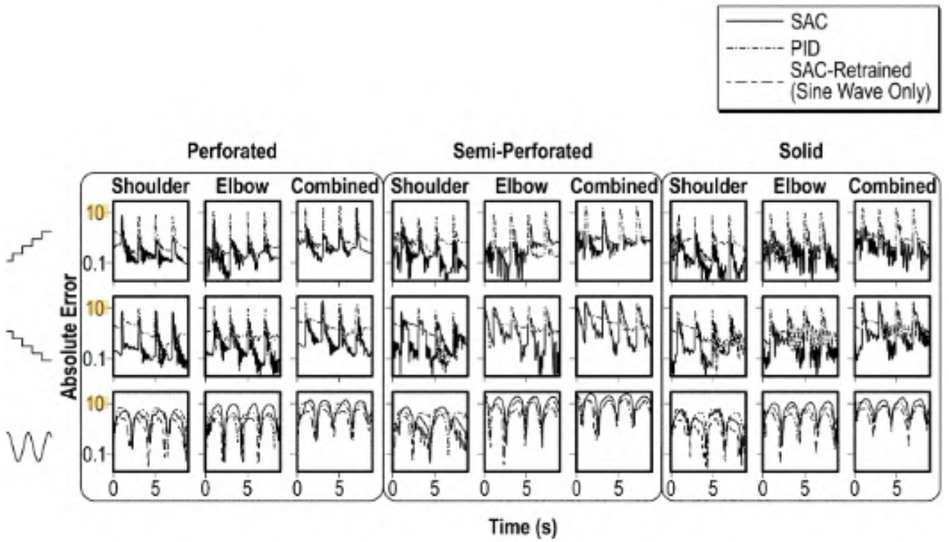


FIG. 27

Appendix 1.6 - PID Plots

- FIG. 28 shows a tuning plot for reaching an angle of about 35° using a PID control algorithm for the shoulder joint
- FIG. 29 shows a 3D visualization grid search values for the shoulder joint for each of the KP, K1, and KD parameters of the PID control algorithm according to the present disclosure
- FIG. 30 shows a tuning plot for reaching an angle of about 35° using a PID control algorithm for the elbow joint
- FIG. 31 shows a 3D visualization grid search values for the elbow joint for each of the KP, K1, and KD parameters of the PID control algorithm according to the present disclosure.

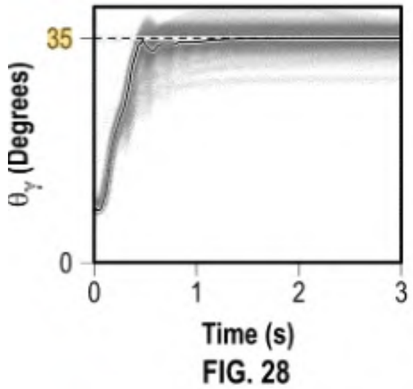


FIG. 28

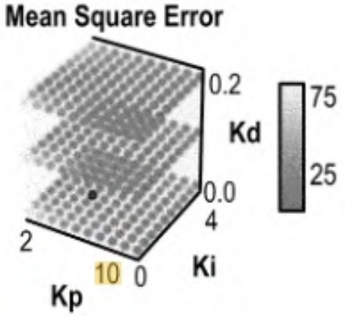


FIG. 29

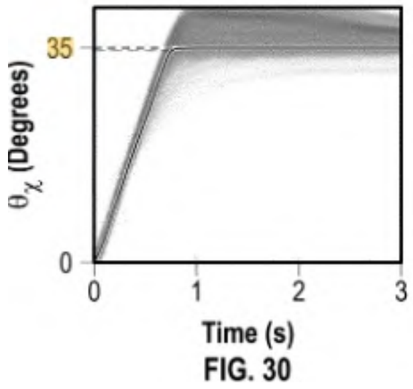


FIG. 30

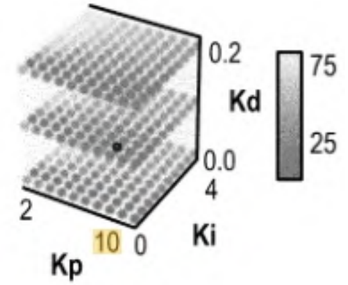
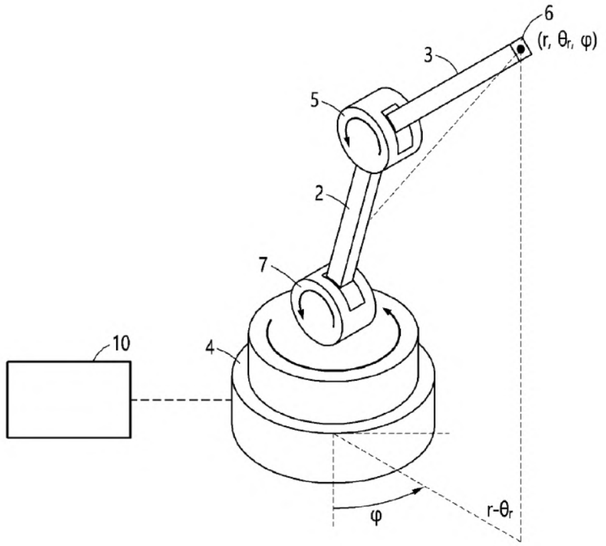


FIG. 31

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Appendix 2.1 - Additional Prior Arts



Robot, Drive Unit For a Robot and Positioning Method

- US 20240001539 A1
- Daegu Gyeongbuk Institute of Science and Technology
- 2023-09-11

Noteable Claims:

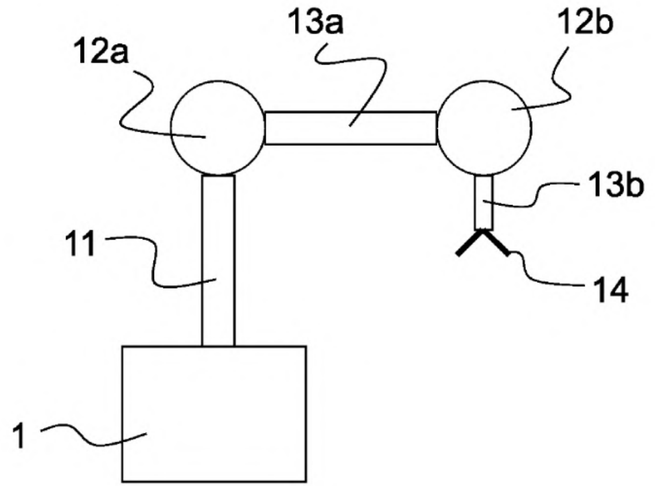
- Uses multidimensional control to correct disturbances
- Handle internal and external disturbances

Differences:

- Does not use machine learning and reinforcd learning

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Appendix 2.2 - Additional Prior Arts



Robust Position Control System of Flexible Joint Robots

- US9342180B2
- Wayne Westerman John G. Elias
- 2009-06-05

Noteable Claims:

- The drive unit with drive shaft
- A method for adjusting angular position of output shaft

Differences:

Is not flexible enough to mimic organic limbs

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Appendix 5 - Sources

Slide 4

- “Artificial Limbs Market to Grow at 5.1% Cagr, Crossing $3.5 Billion by 2031: TMR Study.” Yahoo! Finance, Yahoo!, finance.yahoo.com/news/artificial-limbs-market-grow-5-153000868.html?guccounter=1&guce_referrer=aHR0cHM6Ly93d3cuZ29vZ2xlLmNvbS8&guce_referrer_sig=AQAAAI4Fzt4qnLIHMf_q61wYoLJ5e_q2FAnX8CD6ApCXUtb6n8VNc2ILSbQ402caeGB0OJ14HomlV8Ek9O106J0W-MzShwkGeTuvJNozql6w3F70xCJVvjsvVwSJf5bFWCpykJDPJ5b083OHaH3Bx7iqQpn8jm6DkvVTelgqmHmcBZ4__. Accessed 6 June 2024.
- Marks LJ, Michael JW. Science, medicine, and the future: Artificial limbs. BMJ. 2001 Sep 29;323(7315):732-5. doi: 10.1136/bmj.323.7315.732. PMID: 11576982; PMCID: PMC1121287.

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- University of California, Santa Cruz Industry Alliances & Technology Commercialization. “Ligament-Based Elastic Hybrid Soft-Rigid Joints.” Ligament-Based Elastic Hybrid Soft-Rigid Joints - Available Technology for Licensing from the University of California, Santa Cruz, techtransfer.universityofcalifornia.edu/NCD/33520.html. Accessed 6 June 2024.
- A.S. Robbins, M. Ho, M. Teodorescu, Model-free dynamic control of robotic joints with integrated elastic ligaments, Robotics and Autonomous Systems, Volume 155, 2022, 104150, ISSN 0921-8890, https://doi.org/10.1016/j.robot.2022.104150.
- “US20220281123A1 - Soft-Rigid Robotic Joints Controlled by Deep Reinforcement-Learning.” *Google Patents*, Google, patents.google.com/patent/US20220281123. Accessed 6 June 2024.
- Robbins, A., & Teodorescu, M. (2022, September 8). Soft-rigid robotic joints controlled by deep reinforcement-learning.

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Appendix 5 - Sources

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- University of California, Santa Cruz Industry Alliances & Technology Commercialization. “Ligament-Based Elastic Hybrid Soft-Rigid Joints.” Ligament-Based Elastic Hybrid Soft-Rigid Joints - Available Technology for Licensing from the University of California, Santa Cruz, techtransfer.universityofcalifornia.edu/NCD/33520.html. Accessed 6 June 2024.
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- “US20190261565A1 - Robotic Fruit Picking System.” *Google Patents*, Google, 2020, patents.google.com/patent/US20190261565A1/en?oq=US%2B20190261565%2BA1.
- “US20200061811A1 - Robotic Control System.” *Google Patents*, Google, 2020, patents.google.com/patent/US20200061811A1/en?oq=US%2B20200061811%2BA1.
- “US20220066456A1 - Obstacle Recognition Method for Autonomous Robots.” *Google Patents*, Google, 2019, patents.google.com/patent/US20220066456A1/en?oq=US%2B20220066456%2BA1.

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- “Prosthetics & Orthotics Market - Size, Share & Industry Analysis.” Prosthetics & Orthotics Market - Size, Share & Industry Analysis, www.mordorintelligence.com/industry-reports/prosthetics-and-orthotics. Accessed 6 June 2024.
- Analysis of 1.5% share of market- estimated on other companies in the segment. Ottobock has majority market share with 56%. Others have market share between 5-10% only.

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